

RAVINDU DILSHAN KARUNATHILAKA

Full Stack Developer

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📍 Ambalangoda, Sri Lanka.

PROFESSIONAL SUMMARY

Final-year Computer Science & Technology undergraduate at Uva Wellassa University of Sri Lanka with hands-on experience designing and building full-stack web and desktop applications. Proficient in React.js, Node.js, Express.js, PostgreSQL, and TypeScript. Experienced in RESTful API design, relational and non-relational database management, responsive UI/UX development, and Agile/SDLC workflows. Demonstrated ability to independently architect and deliver production-ready systems from database design through to user interface. Seeking a Software Engineering Internship or Junior Software Engineer role to contribute to scalable, real-world software systems.

TECHNICAL SKILLS

Proficient

JavaScript, TypeScript, PHP, Java, React.js, Next.js, Express.js, Tailwind CSS, CSS3, REST APIs, MySQL, PostgreSQL, MongoDB

Familiar

Python, SpringBoot, C, C#, Vue.js, Angular, Laravel, React Native, Electron.js, Three.js, Socket.IO, .NET, SQLite, Firebase, Docker

Practices

OOP, RESTful API Design, Database Design, Agile/SDLC, Version Control, Debugging

Soft Skills

Analytical Thinking, Problem Solving, Collaborative Development, Communication, Adaptability

Tools & Platforms

Git, GitHub, VS Code, IntelliJ IDEA, Postman, Figma, Jira, Linux

PROJECTS

Pinklet POS - Desktop Point-of-Sale Application · Client Project

Electron.js, React 18, TypeScript, Node.js, Express.js, SQLite, Prisma ORM, Tailwind CSS v4

📅 2026 📍 Pinklet Cakes & Gift Items Shop - Batapola, Sri Lanka.

🔗 <https://github.com/ravindudilshanyk/pinklet-pos>

- Engineered a production-ready offline-first desktop POS system for a real retail bakery business in Sri Lanka, replacing manual billing and paper-based records.
- Architected the application as a pnpm monorepo with 4 packages (Electron, React, Express, shared TypeScript types), ensuring full-stack type safety across the entire codebase.
- Modeled a normalized SQLite database schema with Prisma ORM across 12+ models and established JWT role-based access control (Owner/Cashier) with bcrypt password hashing and email OTP verification.
- Delivered a multi-layer billing engine supporting quick sales and pre-order workflows with simultaneous discount layers, loyalty coin redemptions, and real-time per-item profit calculation.
- Introduced a pre-order lifecycle management system with 5-stage status tracking, advance payment collection, outstanding balance management, and overdue alerts.
- Integrated JS PDF receipt generation in 80mm thermal printer format with dynamic page sizing, itemized discount breakdown, and WhatsApp invoice sharing.
- Produced a business analytics dashboard covering revenue/profit trends, cashier performance tracking, category breakdowns, and inventory reports with configurable date ranges.

PROJECTS

Smart Route LK – Intelligent Bus Booking Platform · Solo Project

React, Node.js, Express.js, PostgreSQL, Socket.IO, Tailwind CSS, JWT, PDFKit, Google OAuth 2.0

📅 2026 - Present 📍 Ambalangoda, Sri Lanka.

🔗 <https://github.com/ravindudilshanyk/smart-route-lk-web>

- Architected a segment-based seat allocation system using a PostgreSQL stop-order overlap detection algorithm, allowing a single seat to serve multiple passengers across different route segments per trip.
- Engineered a connecting bus search engine that discovers two-bus routes with automatic transfer timing validation, returning unified results with combined fare, total journey duration, and transfer points.
- Established dual authentication (JWT + Google OAuth 2.0) with progressive profile completion — users browse freely but are blocked only at booking, mirroring industry-standard UX patterns.
- Structured a 13-table PostgreSQL schema across 4 user roles (Passenger, Bus Owner, Conductor, Admin) with segment-level seat availability tracking and owner payout separation from service fees.
- Generated per-passenger QR tickets with bus-specific validation, enabling conductors to independently verify passengers across multi-leg journeys without cross-bus conflicts.
- Delivered a 4-step bus registration wizard with live seat layout preview, real-time distance-based fare calculation, and atomic database transactions ensuring full data consistency on submission.

Pinklet - Interactive Gift Customization Platform · University Group Project

React.js, .NET, PostgreSQL, Three.js

📅 2025 📍 Uva Wellassa University of Sri Lanka

🔗 <https://github.com/ravindudilshanyk/pinklet>

- Crafted a budget-based gift package suggestion engine where users input occasion, age, and gender preferences and the system automatically recommends curated gift packages across multiple categories.
- Developed a manual package builder allowing customers to edit suggested packages — add, remove, or adjust item quantities — with live pricing updates reflecting changes in real time.
- Introduced a custom cake ordering flow where customers upload a reference image, receive a shop owner price quote, and confirm payment after approval.
- Constructed a 3D cake customization tool using Three.js enabling live visual previewing of cake designs directly in the browser.
- Launched a vendor marketplace enabling local sellers to list products with competitive pricing, expanding available inventory beyond the shop's own stock.
- Spearheaded the complete system design including UI/UX, web flow, and frontend architecture, with active backend contributions delivering end-to-end across Buyer, Vendor, and Admin roles.

Rent IT - Online Rental Platform · University Group Project

React (Vite), PHP, MySQL, PayHere

📅 2024 📍 Uva Wellassa University of Sri Lanka

🔗 <https://github.com/ravindudilshanyk/RentItNew>

- Designed a time-aware rental availability system that automatically blocks item availability during active rental periods, preventing double-booking conflicts.
- Implemented return delivery buffer time logic, ensuring items remain unavailable until safely returned to the owner after each rental period.
- Created an OTP-based item handover and return tracking flow - similar to Uber's delivery model - ensuring verified pickup and return at each stage of the rental lifecycle.
- Developed a damage reporting and accountability system with date extension support requiring additional payment for postponed returns, protecting item owners from loss.
- Integrated PayHere payment gateway for secure online transactions covering rental fees and extension payments.
- Led end-to-end system design including UI/UX, web flow, and frontend architecture, with active backend contributions across Renter, Item Owner, and Platform Admin roles.

RESEARCH PROJECT

Vote SL - A Secure E-Voting Research Prototype

Blockchain, Facial Recognition, Next.js, Node.js, React Native, PostgreSQL

📅 2025 - Present 📍 Uva Wellassa University of Sri Lanka

🔗 <https://github.com/ravindudilshanyk/VoteSL-Mobile>

- Formulated a dual-platform architecture comprising a React Native mobile app for voters and a Next.js admin dashboard for election management, live monitoring, and result publishing.
- Deployed blockchain-based immutable vote storage ensuring tamper-proof audit trails where votes cannot be altered or deleted after submission.
- Incorporated facial recognition biometric authentication to prevent impersonation and eliminate duplicate voting vulnerabilities.
- Structured role-based access control (RBAC) across five roles — Super Admin, Election Officers, Grama Niladhari, Result Officers, and Observers.
- Supports presidential (preference-based) and parliamentary (district-based) election types with a multilingual UI in Sinhala, Tamil, and English.

EDUCATION

Bachelor of Science in Computer Science and Technology

Uva Wellassa University of Sri Lanka

📅 2022 - Present 📍 Badulla, Sri Lanka

CERTIFICATIONS & COURSES

IBM Full Stack Software Developer

Coursera (In Progress)

AI For Everyone, DeepLearning.AI

Coursera (In Progress)

VOLUNTEERING

UI Designer - IEEE Computer Society Official Website

IEEE Student Branch, Uva Wellassa University

- Designed accessible, user-friendly interface layouts for the official IEEE Computer Society chapter website
- Collaborated with the development team to ensure design consistency and seamless implementation across all pages

LANGUAGES

Sinhala	Native	English	Proficient
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REFERENCES

Ms. S. D. H. S. Wickramarathne

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